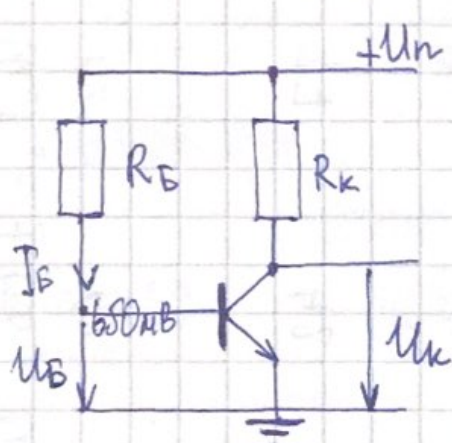


28. Усилитель на биполярных транзисторах.

Цель работы: изучение принципов работы усилителя на биполярных транзисторах, определение параметров усилителя и используемого в нем транзистора.

Ход работы.

1. Нестабилизованный усилитель.



$R_K = 2 \text{ кОм}$. заданные
параметры
 $\hat{I}_K = 1,5 \text{ mA}$.

$$R_B = 620 \text{ кОм}$$

$$U_K = 7,11 \text{ В}$$

$$I_K = \frac{U_n - U_K}{R_K} = \frac{10 \text{ В} - 7,11 \text{ В}}{2 \cdot 10^3 \text{ Ом}} =$$

$$= 1,44 \text{ mA} \sim \hat{I}_K.$$

$$R_B = \frac{U_n - U_B}{I_B} = \frac{(U_n - U_B) h_{21}}{I_K} \quad \text{~~по формуле~~}$$

$$h_{21} = \frac{R_B I_K}{U_n - U_B} = \frac{620 \cdot 10^3 \cdot 1,44 \cdot 10^{-3}}{10 - 660 \cdot 10^{-3}} = 95,6$$

1st group

$$R_k[k_{0u}] = 2$$

$$I_u[u_A] = 1,5$$

$$C_E[u_{\text{ref}}] = 0,068$$

$$u_n = 108$$

Group 1

$$R_E[k_{0u}] = 620$$

$$u_k[B] = 7,118$$

$$u_E[B] = 650 \cdot 10^{-3} \quad h_{21} = 95,5$$

$$u_{\text{box-max}}[B]$$

$$K_u$$

$$K_e$$

$$R_E[u_{0u}]$$

$$f_u[u_u]$$

$$f_E[u_{0u}]$$

Group 2a

$$R_E[u_{0u}] = 620$$

$$2,8$$

$$172$$

$$66$$

$$1,256$$

$$560$$

$$204$$

$$u_k[B] = 5,1$$

$$R_E[u_{0u}] = 1,2$$

$$4,94$$

$$77$$

$$48$$

$$3,411$$

$$510$$

$$265$$

Group 2b

$$u_k[B] = 7,5$$

$$R_E[u_{0u}] = 620$$

$$1,44$$

$$91$$

$$34$$

$$1,178$$

$$495$$

$$424$$

Group 3
 $C_E[u_{\text{ref}}] = 220$

$$R_E[u_{0u}] = 2$$

Group 4

$$R_E[u_{0u}] = 56$$

$$u_k[B] = 9,4$$

$$R_E[u_{0u}] = 43$$

$$u_E[B] = 0,68$$

$$R_E[u_{0u}] = 0,17$$

$$u_k[B] = 0,09$$

Group 5

$$C_E[u_{\text{ref}}] = 220$$

$$350,4 \cdot 10^{-3}$$

$$16$$

$$7$$

$$1,82$$

$$4,1 \cdot 10^3$$

$$408$$

Group 6

$$T_u[u_c] = 0,369$$

$$C_E[u_{\text{ref}}] = 220$$

$$T_E[u_c] = 3,809$$

$$\frac{1}{4(2\pi f_u)[B]} = \frac{1}{4(2\pi f_E)[B]} =$$

$$= 431$$

$$= 44,8$$

Group 7

$$80,5 \cdot 10^{-3}$$

$$3,13$$

$$1,88$$

$$3$$

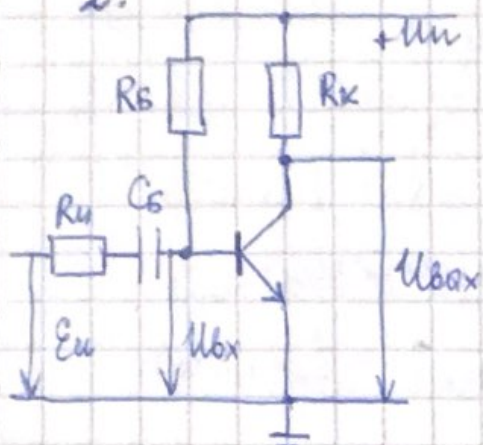
$$15 \cdot 10^3$$

$$758$$

Group 8

qbyxioacupum
ycunufek

2.



$$R_u \approx R_K = 2 \text{ k}\Omega$$

$$a) R_B = 620 \text{ k}\Omega$$

$$U_{bax_max} = 2,8 \text{ B}$$

$$f_H = 560 \text{ Гц}, \quad f_B = 204 \text{ кГц}$$

$$U_{bx} = 12,75 \text{ мВ}$$

$$U_{bax} = 2,19 \text{ B}$$

$$E_u = 33,05 \text{ мВ}$$

$$K_u = \frac{U_{bax}}{U_{bx}} = 172, \quad K_e = \frac{U_{bax}}{E_u} = 66$$

$$R_{bx} = \frac{U_{bx}}{I_{bx}} = U_{bx} \cdot \frac{R_u}{E_u - U_{bx}} = 1,256 \text{ k}\Omega$$

b) ~~Решение~~

$$U_{bax_max} = \cancel{1,802 \text{ B}} 4,94 \text{ B}$$

$$f_H = \cancel{510 \text{ Гц}}, \quad f_B = \cancel{265 \text{ кГц}}$$

$$U_{bax} = \cancel{1,56 \text{ B}} 4,15 \text{ B}$$

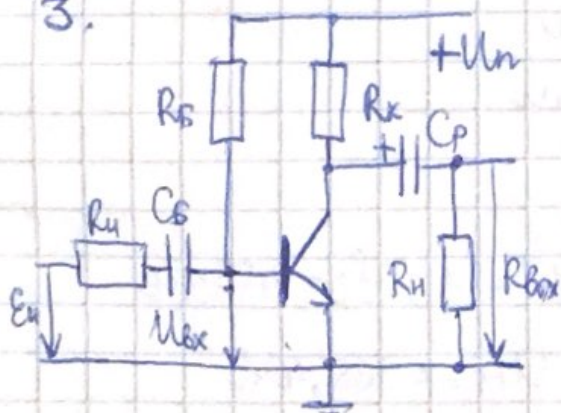
$$U_{bx} = \cancel{9,12 \text{ мВ}} 58 \text{ мВ}$$

$$E_u = \cancel{33,36 \text{ мВ}} 92 \text{ мВ}$$

$$K_u = \cancel{77}, \quad K_e = 48$$

$$R_{bx} = \cancel{1,617 \text{ k}\Omega} 3,4 \text{ k}\Omega$$

3.



$$R_u \approx R_K = 2 \text{ k}\Omega$$

$$R_u \approx R_K$$

$$C_p = 220 \text{ мкФ}$$

$$U_{bax_max} = 1,44 \text{ B}$$

$$U_{bax} = 1,123 \text{ B}$$

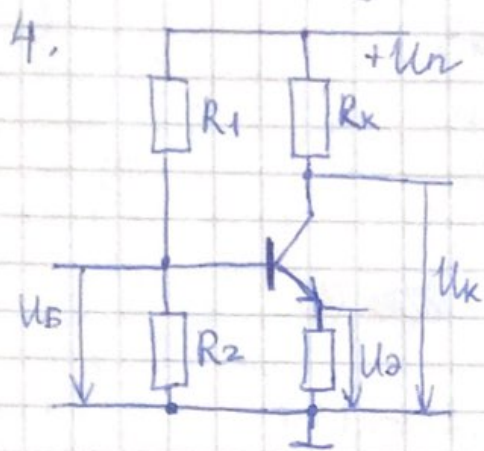
$$U_{bx} = 12,39 \text{ мВ}$$

$$E_u = 33,42 \text{ мВ}$$

$$K_u = 91, K_e = 34, R_{bx} = 1,178 \text{ кОм}.$$

$$f_H = 495 \text{ Гц}, f_B = 424 \text{ кГц}.$$

Стабилизированный усилитель.



$$R_2 = 470 \text{ Ом} \quad \hat{I}_k = 1,5 \text{ mA}.$$

$$R_2 \approx 2R_k = 4,3 \text{ кОм}$$

$$I_k \approx \frac{E_B - 0,6 \text{ В}}{R_2}$$

$$\rightarrow \hat{E}_B = 0,7 \text{ В}.$$

$$E_B = U_b \frac{R_2}{R_1 + R_2}, \quad \hat{R}_1 = \frac{U_b R_2}{\hat{E}_B} - R_2 \approx 57 \text{ кОм}$$

$$R_1 = 56 \text{ кОм}, \quad E_B = 0,68 \text{ В}, \quad I_k \approx 1,44 \text{ mA}.$$

~~UB~~ $E_B = 0,68 \text{ В}.$

$$U_o = 0,09 \text{ В}$$

$$U_k = 9,4 \text{ В}.$$

5. $C_3 = 220 \text{ мФ}$ $R_1 = 56 \text{ кОм}$

$$R_k \approx R_c = 2 \text{ кОм} \quad R_2 = 4,3 \text{ кОм}.$$

$$C_5 = 0,068 \text{ мкФ}$$

$$U_{вх_макс} = 350,4 \text{ мВ} \quad f_H = 4,1 \text{ МГц} \quad f_B = 408 \text{ кГц}$$

$$U_{вх} = 17,24 \text{ мВ}, \quad E_u = 36,2 \text{ мВ}, \quad U_{вх} = 268,4 \text{ мВ}.$$

$$K_u = \frac{268,4}{17,24} = 16, \quad K_e = 7.$$

$$R_{bx} = 1,8 \text{ k}\Omega$$

$$6. a) T \approx 20 \frac{1}{2\pi f_H} \rightarrow f = 1,3 \text{ kHz}$$

$$\Delta U = 1,261 \text{ B} \approx 1U_0$$

$$0,37U_0 = 0,4665 \text{ B}$$

$$T_H = 368,9 \text{ мкс.}$$

$$b) T \approx 20 \frac{1}{2\pi f_B} \rightarrow f = 1,28 \text{ kHz.}$$

$$\Delta U = 1,904 \text{ B} \approx 2U_0$$

$$0,37U_0 = 0,704 \text{ B}$$

$$T_B = 3,809 \text{ мкс.}$$

$$7. C_3 = 0.$$

$$U_{\text{вых-макс}} = 80,5 \text{ мВ}$$

$$f_H = 15 \text{ kHz}, f_B = 758 \text{ kHz}$$

$$U_{\text{вых}} = 68,11 \text{ мВ}, U_{\text{bx}} = 21,74 \text{ мВ}, E_H = 36,22 \text{ мВ}$$

$$K_u = \frac{68,11}{21,74} = 3,13.$$

$$K_e = \frac{68,11}{36,22} = 1,88.$$

$$R_{bx} = 3 \text{ k}\Omega.$$